

THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION, NOVEMBER 2021**Mathematics****MEC 3C 03—MATHEMATICAL ECONOMICS****(2019—2020 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 3C 03—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. When $Y''(t) = 10$, $Y(t)$ will be :
- (A) 1. (B) 0.
(C) $5t^2$. (D) $5t^2 + tc_1 + C$.
2. What is the degree of $\left(\frac{dy}{dx^2}\right)^6$:
- (A) 2. (B) 3.
(C) 4. (D) 6.
3. Curve that is also known as equal product curve is :
- (A) Indifference Curve. (B) Isoquants.
(C) Demand Curve. (D) None of the above.
4. The order of $I_t = a(y_{t-1} - y_{t-2})$ is :
- (A) 1. (B) -1.
(C) 2. (D) -2.
5. When $\sigma = 0$, substitution will be :
- (A) Possible. (B) Sometimes possible.
(C) Impossible. (D) Cannot say.
6. The third stage in the law of variable proportion is called :
- (A) Increasing returns. (B) Diminishing returns.
(C) Negative returns. (D) Proportional return.
7. Given $\frac{\partial Q}{\partial x_1}$, where x_1 is input represents :
- (A) AP_{x_1} . (B) MP_{x_1} .
(C) MRTS. (D) MRS.

8. When the total product is maximum, marginal product will be ?
- (A) Minimum. (B) Zero.
(C) Maximum. (D) Negative.
9. When demand for a good is give by $Q = 40 - P$, the maximum amount that would be demanded at nil price is :
- (A) 1. (B) 0.
(C) 40. (D) 400.
10. Technical relationship between input and output is called :
- (A) Elasticity. (B) Production function.
(C) Input function. (D) None of the above.
11. Higher isoquants represents higher :
- (A) Profit. (B) Output.
(C) Cost. (D) None of the above.
12. What is the degree of $\left(\frac{d^3y}{dx^3}\right)^4 + \left(\frac{d^2y}{dx^2}\right)^6 = 10 - Y$:
- (A) 3. (B) 4.
(C) 2. (D) 6.
13. The producer will be in equilibrium when :
- (A) $MRTS = \frac{P_x}{P_y}$. (B) $MRTS > \frac{P_x}{P_y}$.
(C) $MRTS < \frac{P_x}{P_y}$. (D) None of the above.
14. Harred model explains ————— growth of the economy.
- (A) Static. (B) Dynamic.
(C) Equilibrium. (D) Balanced.

15. A line that connects various equilibrium points of producer is :
- (A) Isocost line. (B) Isoquants.
(C) Expansion path. (D) Price line.
16. Cobb Douglas production function is of degree :
- (A) One. (B) Two.
(C) Three. (D) Four.
17. Iso quants are downward sloping and ————— to the origin.
- (A) Convex. (B) Concave.
(C) Vertical. (D) Horizontal.
18. What is the order of $\frac{d^3y}{dx^3} + (x^2Y)\frac{d^2y}{dx^2} - 4Y^4 = 0$:
- (A) First. (B) Second.
(C) Third. (D) Fourth.
19. MRTS is the slope of :
- (A) Production function. (B) Priceline
(C) Isocostline. (D) Isoquant.
20. For producer which is rational stage for producer in law of variable proportion :
- (A) First. (B) Second.
(C) Third. (D) None of the above.

THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION, NOVEMBER 2021

Mathematics

MEC 3C 03—MATHEMATICAL ECONOMICS

(2019—2020 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.
Each question carries 3 marks.
All questions can be attended.
Overall Ceiling 24.*

1. Define difference equation with an example.
2. Write down the standard form of a first order linear differential equation. What is the general equation used to solve a first order linear difference equation ?
3. Distinguish between order and degree of differential equations with example.
4. What is Marginal Rate of Technical Substitution ?
5. Write the first order partial derivatives for the function $q = 5 K^{0.4} L^{0.6}$.
6. What do you mean by elasticity of substitution ?
7. Explain the properties of Cobb-Douglas Production Function.
8. What are returns to scale ?
9. What is Certainty-Equivalent approach ?
10. Explain Law of Variable Proportions.
11. What do you mean by a Decision Tree ?
12. State briefly the properties of isoquants.

(8 × 3 = 24 marks)

Section B

*Answer at least five questions.
Each question carries 5 marks.
All questions can be attended.
Overall Ceiling 25.*

13. Solve the differential equation $\frac{dy}{dx} + \frac{1}{x}y = x$.
14. Solve the difference equation $3y_{x+1} = 3y_x - 7$.

Turn over

15. Explain Economic Regions of Production Using Ridge Lines.
16. How will you optimize Cobb-Douglas production Function ?
17. Discuss the Probability Distribution approach.
18. What do you mean by solution of a difference equation ? Distinguish between general solution and particular solution of differential equations.
19. Explain producer's Equilibrium with graphical representation.

(5 × 5 = 25 marks)

Section C

Answer any one question.

The question carries 11 marks.

20. What do you mean by Investment Appraisal ? Explain Various Investment Appraisal methods.
21. Discuss how differential equations are used in the area of Economics.

(1 × 11 = 11 marks)